

ORIGINAL ARTICLE

Follistatin Alleviates Vascular Dysfunction by Reducing Oxidative Stress and Promoting Browning in Perivascular Adipose Tissue

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BACKGROUND: Essential hypertension, characterized by vascular dysfunction, remains a leading modifiable cause of death globally. Perivascular adipose tissue (PVAT), which normally reduces vasoconstriction, becomes contractile in hypertension due to increased reactive oxygen species (ROS). Brown PVAT has emerged as a protective regulator of vascular tone. Follistatin, an activin antagonist, induces adipose tissue browning via AMPK (AMP-activated protein kinase). We recently reported improved blood pressure and vascular function by follistatin in the spontaneously hypertensive rat (SHR) model of essential hypertension. Here, we investigate whether follistatin reduces ROS and induces browning in SHR PVAT to restore PVAT-mediated vascular function.

METHODS: SHRs were treated with vehicle, follistatin, or hydralazine for 8 weeks. Mesenteric white and thoracic brown PVAT from SHR and normotensive Wistar-Kyoto mice were used, with wire myography used to assess vascular function. PVAT nitric oxide, ROS, and browning markers were assessed via immunohistochemistry. PVAT was treated ex vivo to assess mechanisms of browning. Unbiased proteomic analysis of PVAT was performed using liquid chromatography-mass spectrometry.

RESULTS: SHR PVAT dysfunction, manifesting as a procontractile effect, was inhibited by follistatin through reducing ROS, enhancing NO bioavailability, and promoting browning. These effects were mediated by activin neutralization and AMPK phosphorylation. Hydralazine-induced blood pressure reduction did not replicate follistatin effects, suggesting vascular benefits of follistatin are mediated by direct PVAT modulation. Proteomic analysis revealed a shift in proteome towards a normal state, upregulating processes associated with adipose browning.

CONCLUSIONS: Follistatin restores PVAT-mediated vascular relaxation via ROS reduction, activin neutralization, and AMPK-dependent browning, positioning the potential for PVAT as a therapeutic target for vascular dysfunction in essential hypertension. (*Hypertension*. 2026;83:00-00. DOI: 10.1161/HYPERTENSIONAHA.126.26722.)

• **Supplement Material.**

Key Words: activin ■ essential hypertension ■ follistatin ■ myography ■ oxidative stress

The contribution of perivascular adipose tissue (PVAT) in hypertension is becoming increasingly recognized. PVAT surrounds most blood vessels, exerting a paracrine effect through the release of vasoactive substances. In normal conditions, PVAT exerts an anticontractile effect. In hypertension, PVAT exhibits increased oxidative stress resulting in the augmented

release of contractile factors.¹ Notably, PVAT dysfunction precedes the development of hypertension in the spontaneously hypertensive rat (SHR) model of essential hypertension,² suggesting a potential role in its pathophysiology.

Adipose tissue is heterogeneous and broadly classified as either white, brown, or beige. White adipose

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NOVELTY AND RELEVANCE

What Is New?

Mechanisms underlying the contribution of white perivascular adipose tissue (PVAT) to resistance artery function in essential hypertension are incompletely understood. This is the first study to demonstrate that the activin antagonist, follistatin, inhibits oxidative stress and induces browning of white PVAT around resistance vessels to alleviate vascular dysfunction in the spontaneously hypertensive rat model of essential hypertension.

What Is Relevant?

Our study reveals that PVAT around mesenteric resistance vessels in spontaneously hypertensive rats

exhibits excessive oxidative stress and a reduced brown phenotype, which is associated with vascular oxidative damage and vasoconstriction. Augmented activin signaling and reduced AMPK (AMP-activated protein kinase) signaling contribute to the impaired anticontractile effect of PVAT.

Clinical/Pathophysiological Implications?

Follistatin-mediated inhibition of activin signaling may represent a novel therapeutic strategy to restore the anticontractile effect of PVAT on resistance vessel function, thereby attenuating elevated blood pressure in essential hypertension.

Nonstandard Abbreviations and Acronyms

AMPK	AMP-activated protein kinase
BAT	brown adipose tissue
BP	blood pressure
ERK	extracellular signal-regulated kinase
HBSS	Hank's balanced salt solution
NO	nitric oxide
NOX	NADPH oxidases
p38 MAPK	p38 Mitogen-activated protein kinase
PRDM16	PR domain-containing 16
PVAT	perivascular adipose tissue
ROS	reactive oxygen species
SHR	spontaneously hypertensive rat
SHR	spontaneously hypertensive rat
Smad	small mothers against decapentaplegic
TGF-β	transforming growth factor β
UCP-1	uncoupled protein 1
WAT	White adipose tissue
WKY	Wistar Kyoto

tissue (WAT) accumulates in metabolic diseases such as obesity and is characterized by excessive oxidative stress and inflammation.³ In contrast, brown adipose tissue (BAT) has a more potent anticontractile effect and protects against obesity-induced inflammation and vascular dysfunction.⁴ The presence of BAT is correlated with a lower prevalence of hypertension in individuals.⁵ On β₃-adrenergic receptor activation or cold exposure, WAT undergoes a phenotypic transformation into BAT, a process referred to as adipose tissue browning. PVAT browning has been associated with blood pressure (BP) reduction.⁶

We recently found that the glycoprotein follistatin significantly lowers BP and improves vessel function through the inhibition of oxidative stress in essential and secondary hypertension models.⁷ Follistatin inhibits several members of the TGF-β (transforming growth factor β) superfamily with potent effects against activins A and B.⁸ Activins have been implicated in the pathophysiology of hypertension through the regulation of oxidative stress and vasoconstriction.^{9,10} Activin A has also been shown to suppress the differentiation of brown adipocytes.¹¹ Conversely, follistatin induces brown adipocyte differentiation in vitro¹² and the browning of peripheral WAT in vivo^{13,14} via AMPK (AMP-activated protein kinase). Whether follistatin induces PVAT browning is unknown.

In this study, we investigated the effects of follistatin on PVAT structure and function in the SHR model of essential hypertension. We demonstrate that follistatin significantly improves PVAT-mediated vascular function by reducing reactive oxygen species (ROS) and promoting PVAT browning via activin neutralization and AMPK phosphorylation. These effects are not replicated by BP lowering alone with hydralazine. Together, these findings highlight novel mechanisms by which follistatin restores vascular homeostasis and support the therapeutic potential of targeting PVAT in hypertension.

METHODS

The authors declare that all supporting data are available within the article and its online supplementary files. A detailed Methods section is provided in [Supplemental Material](#).

Animal Studies

Experiments adhered to guidelines set by McMaster University and the Canadian Council on Animal Care and were approved by the McMaster University Animal Research Ethics Board.

Research also adhered to ARRIVE guidelines. Twelve-week-old male SHRs and Wistar Kyoto (WKY, normotensive control) were used. Both rat strains were sourced from our in-house breeding colony. First-order mesenteric arteries and PVAT from chronically treated rats are from our previously published work.⁷ Rats were treated intraperitoneally every other day for 8 weeks as follows: WKY Vehicle (PBS, $n=10$), SHR Vehicle (PBS, $n=10$), and SHR FST (follistatin-288, 0.078 mg/kg, $n=10$). Previous mouse studies assessed the effects of different doses and dosing frequencies of follistatin-288.^{7,15} An intermediate dose (0.15 mg/kg) given every other day intraperitoneally—shown to be most effective in lowering BP and improving kidney function in mice—was selected and converted for its use in rats using published guidelines.¹⁶ In a second study, SHRs aged 12 weeks were treated with hydralazine (10 mg/kg per day) in drinking water for 8 weeks. The dose was selected based on its efficacy to reduce BP to a similar level as follistatin (20 mm Hg reduction). All animals were weighed, and appearance and body condition were assessed every other day before treatment administration.

Statistical Analysis

Data were analyzed using GraphPad Prism 11. One-way ANOVA was used to assess statistical significance of BP and histology data. Weekly BP data were analyzed using a repeated measures test. Myograph data were analyzed using a nonlinear regression model. Two-way ANOVA was used to compare differences between 2 curves. Imaging data were evaluated using 1-way ANOVA with a posthoc Tukey correction for multiple comparisons. A Student *t* test was used to compare 2 groups; a normal distribution and homoscedasticity were confirmed using an Anderson-Darling test and *F*-test, respectively. A simple linear regression analysis was performed to evaluate the relationship between PVAT browning and anticontractile effects. Proteomic data were analyzed using Proteome Discoverer with the *Rattus norvegicus* fasta database. Gene Ontology for biological processes was performed on differentially expressed proteins using the Database for Annotation Visualization and Integrated Discovery. The null hypothesis was rejected for *P* values <0.05 . Data are presented as mean $\pm$ SEM, with the number of samples (*n*) indicated in figure captions or as individual data points in the graphs.

RESULTS

Follistatin Prevents the Procontractile Effect of PVAT in SHRs

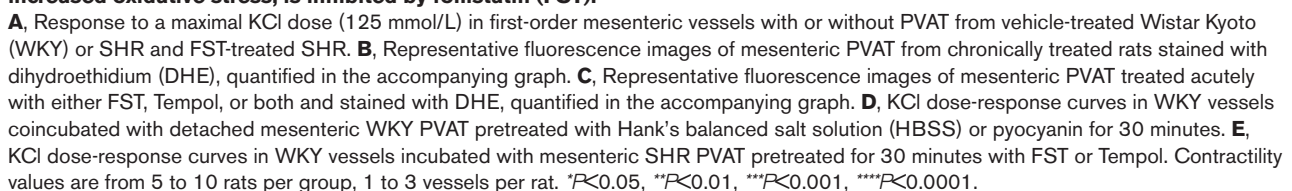
We first determined the effect of PVAT on smooth muscle contraction to KCl in first-order mesenteric arteries isolated from SHRs treated for 8 weeks with vehicle or follistatin, or normotensive rats treated for 8 weeks with vehicle. PVAT exhibited a significant anticontractile effect on WKY vessels (Figure 1A). In contrast, SHR PVAT elicited a marked procontractile response which was abolished by follistatin. The lack of an anticontractile effect by SHR PVAT was also seen in response to the α_1 adrenergic receptor agonist phenylephrine (Figure S1). Conversely, follistatin decreased contraction

to phenylephrine in vessels with PVAT. To eliminate the vasoactive contribution of the endothelium, we assessed KCl responses in endothelium-denuded SHR vessels with or without PVAT. Endothelium removal in SHR vessels augmented contraction to KCl (Figure S2A), suggesting that endothelium-derived factors modulate vascular tone in the SHR. Notably, the procontractile effect of denuding the endothelium was further exacerbated in the presence of SHR PVAT, with PVAT effects reversed by follistatin (Figure S2B). These data highlight a PVAT-specific effect of follistatin with no effect seen on the smooth muscle.

PVAT ROS Induces Vascular Oxidative Stress and Vasoconstriction

Our previous study showed that follistatin inhibits ROS in SHR vessels.⁷ We next investigated whether follistatin exerts similar antioxidant effects in PVAT. Dihydroethidium fluorescence imaging showed increased ROS in SHR PVAT, which was reduced by follistatin (Figure 1B). To better understand whether follistatin acts directly on PVAT, SHR PVAT was treated ex vivo with follistatin for 30 minutes. As shown in Figure 1C, follistatin and the ROS inhibitor Tempol both reduced dihydroethidium fluorescence to similar levels, without an additive effect. To determine whether PVAT could induce vascular ROS, we performed PVAT transfer experiments. PVAT obtained from an SHR treated for 8 weeks with vehicle or follistatin was incubated with a donor vessel obtained from an untreated SHR, thereby eliminating any direct effects of chronic follistatin treatment on the vessel itself (Figure S3A). After a 30-minute incubation, PVAT from vehicle-treated SHRs increased vascular ROS, measured by dihydroethidium. This was not seen in vessels incubated with PVAT from follistatin-treated SHRs (Figure S3B). These findings suggest that PVAT-derived factors induce vascular oxidative stress in SHRs, an effect inhibited by follistatin.

We next examined whether ROS altered vessel contraction. Indeed, pretreatment of WKY vessels with the ROS inducer pyocyanin increased constriction to KCl (Figure S3C). To isolate PVAT effects, we assessed response to KCl in WKY vessels incubated with detached PVAT that had been treated ex vivo with pyocyanin. Pyocyanin treatment of PVAT augmented WKY vessel contraction to KCl (Figure 1D). Conversely, the treatment of detached SHR PVAT with either follistatin or Tempol reduced contraction in WKY vessels following incubation (Figure 1E). Collectively, these findings highlight the potential of PVAT ROS to directly influence vascular tone and demonstrate that the augmented vasoconstriction induced by SHR PVAT is driven at least in part by oxidative stress that is prevented by follistatin.



Activin Upregulation Contributes to Increased ROS in SHR PVAT

The follistatin-288 isoform used in this study is expected to deposit in tissue through its affinity for heparan sulfate proteoglycans.¹⁷ We therefore characterized the abundance of follistatin and activins in PVAT by immunohistochemistry. Although follistatin was significantly reduced in SHR compared with WKY PVAT, exogenous follistatin administration to SHR exceeded levels seen in WKY PVAT (Figure 2A). Concordantly, the elevated expression of activins A and B in SHR PVAT was reduced by follistatin (Figure 2B and 2C).

We previously demonstrated increased vascular ROS on acute treatment (30 minutes) with activin A.⁷ To determine whether activins could induce ROS in PVAT, we treated WKY PVAT with activin A or B. As shown in Figure 2D, both activin A and B increased ROS as measured by dihydroethidium. Furthermore, an activin receptor blocker, SB-431,542, lowered ROS in SHR PVAT, demonstrating the contribution of activin signaling to PVAT oxidative stress in the SHR (Figure 2E).

Vascular ROS are primarily produced by NOX (NADPH oxidases).¹⁸ Among the various NOX isoforms, NOX4 is abundantly expressed in all parts of the vessels, predominantly producing hydrogen peroxide (H_2O_2).¹⁸ We thus sought to identify the contribution of H_2O_2 to the dihydroethidium signal using Amplex Red. SHR PVAT were found to have elevated H_2O_2 levels compared with WKY PVAT, which were reduced by follistatin, but not by heat-denatured follistatin (Figure 2F). These data indicate that the tertiary structure of follistatin is required for its inhibition of ROS. Activin A or B neutralization also significantly reduced H_2O_2 levels in SHR PVAT, supporting their role as important mediators of ROS production in SHR PVAT. To identify a direct role for activin A in ROS production and assess whether this is mediated by NOX, we assessed ROS levels in WKY PVAT treated with activin A in the presence of a NOX1/4 dual inhibitor, GKT137831. The activin A-induced increase in dihydroethidium was prevented by cotreatment with GKT137831 (Figure S4). This finding suggests that ROS production in PVAT is mediated by activin A through NOX activity, and follistatin may reduce PVAT ROS through activin neutralization.

Follistatin Increases NO Bioavailability in SHR PVAT

Superoxide generated by NADPH oxidase reacts with nitric oxide (NO), thereby reducing NO bioavailability and contributing to increased vasoconstriction in hypertension.^{19,20} Peroxynitrite, a product of superoxide and NO, reacts with tyrosine residues in proteins to form nitrotyrosine. Detection of nitrotyrosine by immunohistochemistry

was significantly increased in SHR compared with WKY PVAT, and this was reduced by follistatin (Figure 3A). Further, follistatin increased expression of the primary vascular NO producer, eNOS (endothelial NO synthase; Figure 3B). To determine if follistatin altered NO bioavailability, we used the 4-amino-5-methylamino-2'-7'-difluorofluorescein fluorescence probe to directly detect NO in PVAT. On addition of the NO donor, NONOate, SHR PVAT showed significantly reduced fluorescence intensity compared with WKY PVAT. This indicates that NO is rapidly taken up in SHR PVAT, likely by ROS (Figure 3C). Pretreatment of SHR PVAT with follistatin for 30 minutes increased fluorescence, indicating an increase in NO bioavailability. Tempol restored NO bioavailability to normal levels in SHR PVAT, supporting a key role for oxidative stress in its reduction.

We next assessed whether H_2O_2 reduced NO in PVAT. Under physiological conditions, NO is rapidly oxidized to nitrite, measured fluorometrically by the Griess reagent. Figure 3D shows that follistatin, but not heat-denatured follistatin, increases nitrite production in SHR PVAT. These data suggest that the biological function of follistatin is required for its increase in NO availability. In keeping with this, activins A and B reduce nitrite levels in WKY PVAT (Figure 3E). Conversely, activin A or B neutralization increases nitrite production in SHR PVAT (Figure 3F), highlighting a role for activins in regulating NO, likely via ROS.

Follistatin Induces Adipose Tissue Browning

Adipose tissue browning is associated with improved vasorelaxation and more recently, lower BP.⁶ Follistatin has been shown to induce browning in peripheral adipose depots in both *in vitro* and *in vivo* models, although its effects on PVAT have not been examined. We thus assessed whether follistatin induces browning of white mesenteric PVAT. We measured adipocyte size, which is smaller in BAT compared with WAT.²¹ Figure 4A shows that adipocytes in the SHR white PVAT were enlarged compared with WKY PVAT, and this was normalized by follistatin. Follistatin also increased expression of the BAT markers PRDM16 (PR domain-containing 16; Figure 4B) and UCP-1 (uncoupled protein 1; Figure 4C). In brown aortic PVAT, the downregulation of UCP-1 in SHR was reversed by follistatin (Figure S5A). These data demonstrate the ability of follistatin to induce browning in both phenotypically white and brown PVAT. We confirmed that follistatin also induced browning of inguinal WAT, a peripheral WAT depot, evidenced by increased PRDM16 (Figure S5B).

We next assessed differential effects of white and brown PVAT on vessel response to KCl. WKY vessels were incubated with detached white mesenteric or brown aortic PVAT, and constriction to KCl was assessed. As shown in Figure 4D, brown PVAT induced a more potent

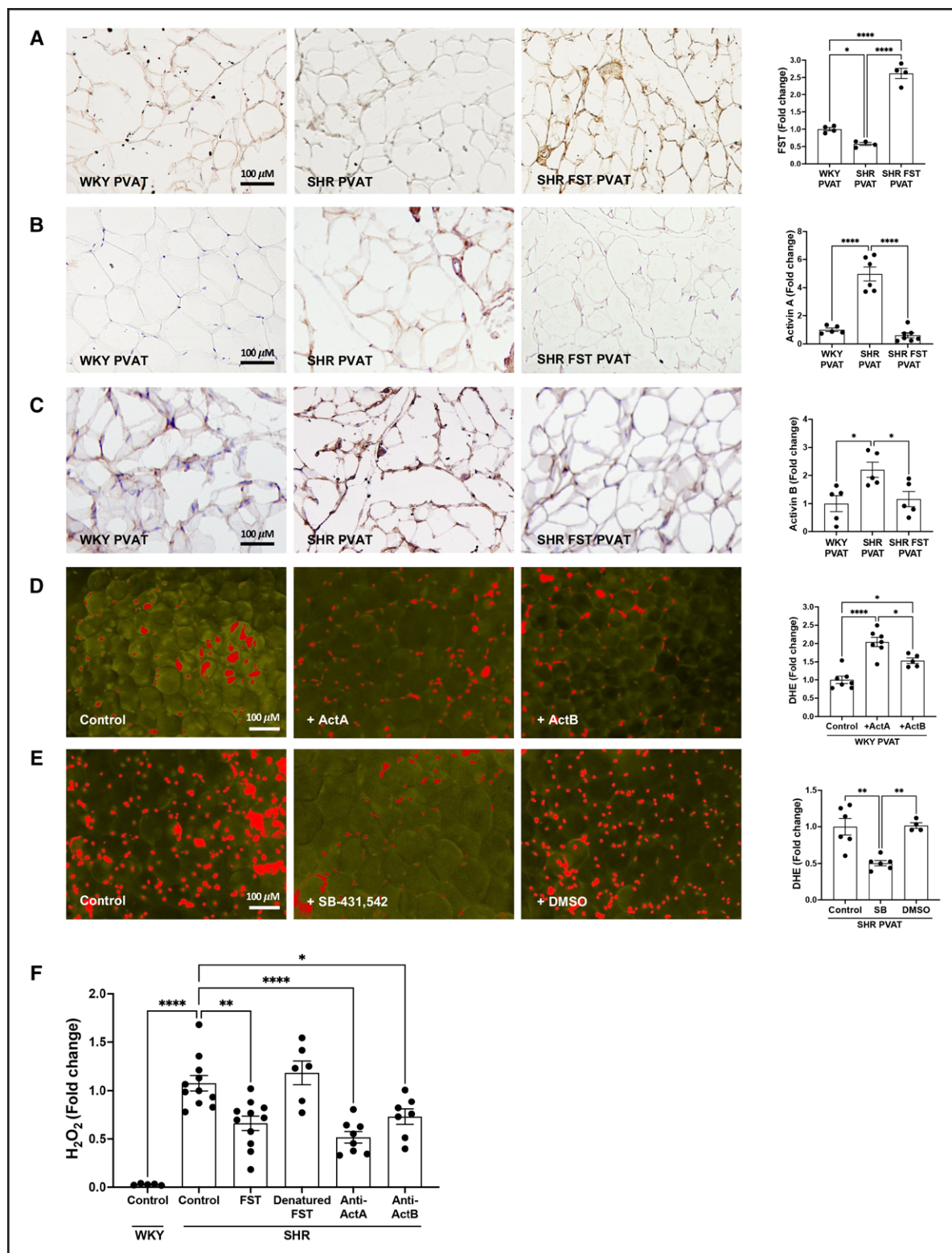


Figure 2. Follistatin (FST) inhibits oxidative stress in perivascular adipose tissue (PVAT) by neutralizing actives.

Representative cross-sectional images of mesenteric PVAT from chronically treated rats stained for (A) FST, (B) actin A, (C) actin B, quantified in accompanying graphs. D, Representative fluorescence images of Wistar Kyoto (WKY) mesenteric PVAT treated acutely with actin A or B for 30 minutes. E, Representative fluorescence images of spontaneously hypertensive rat (SHR) mesenteric PVAT treated acutely with actin receptor blocker SB-431 542 or DMSO for 30 minutes, quantified in accompanying graphs. F, Fold change in H₂O₂ levels assessed using Amplex Red measured in mesenteric PVAT from WKY or SHR treated for 30 minutes with either Hank's balanced salt solution (HBSS), FST, heat-denatured follistatin, or neutralizing antibodies to actin A or actin B. **P*<0.05, ***P*<0.01, *****P*<0.0001.

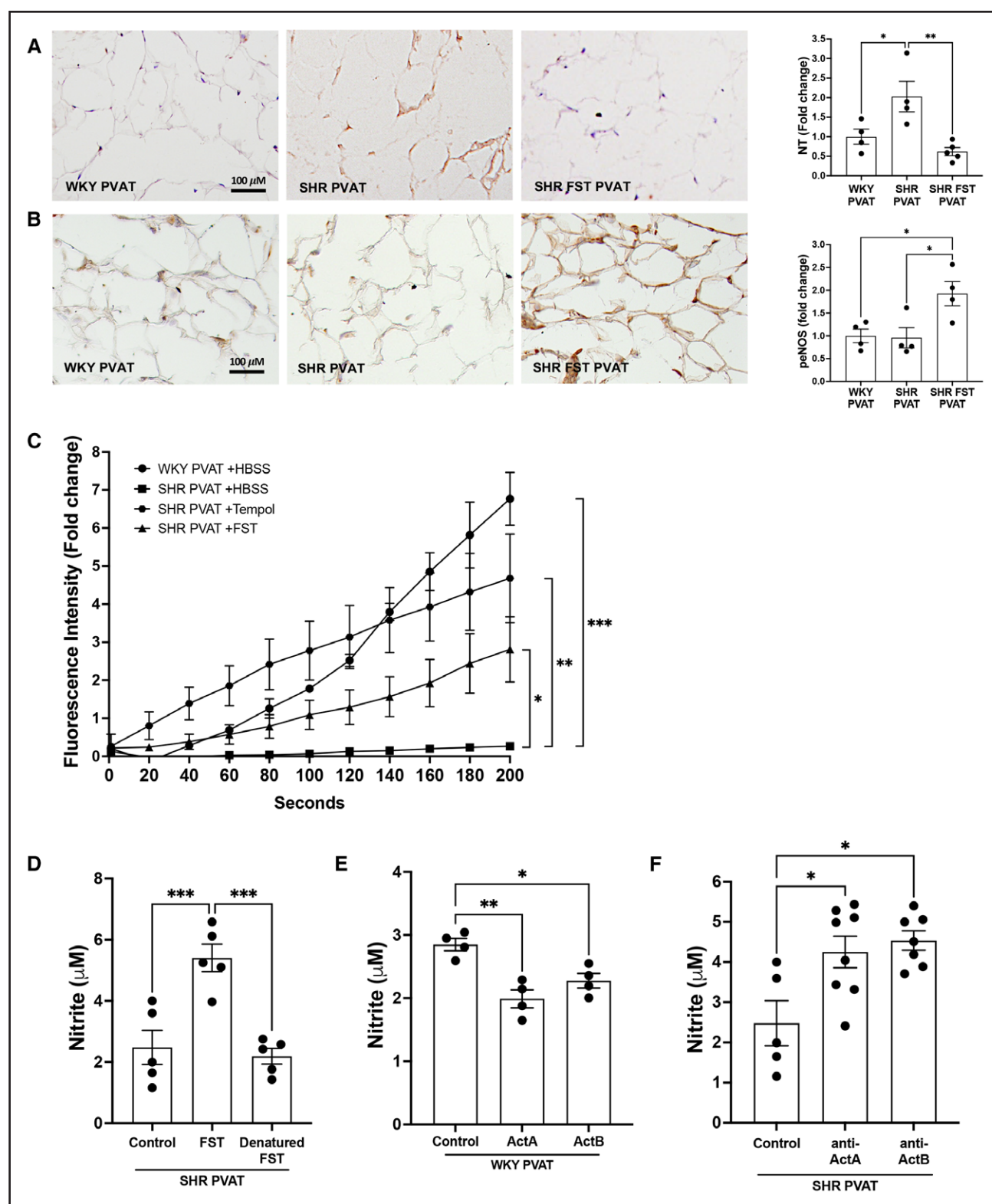


Figure 3. Follistatin (FST) increases NO bioavailability in spontaneously hypertensive rat (SHR) perivascular adipose tissue (PVAT).

Representative cross-sectional images of mesenteric PVAT from chronically treated rats stained for (A) nitrotyrosine and (B) phosphorylated eNOS, quantified in the accompanying graphs. C, Fluorescence intensity measured over a 200-second period in mesenteric PVAT treated for 30 minutes with Hank's balanced salt solution (HBSS), FST, or Tempol, stained with 4-amino-5-methylamino-2'-7'-difluorofluorescein. D, Nitrite levels measured in lysate of SHR PVAT treated for 30 minutes with FST or heat-denatured FST using the Griess reagent kit. Nitrite levels measured in lysate of E. Wistar Kyoto (WKY) PVAT treated for 30 minutes with activin A or B or F. SHR PVAT treated for 30 minutes with neutralizing antibodies to activin A or B. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

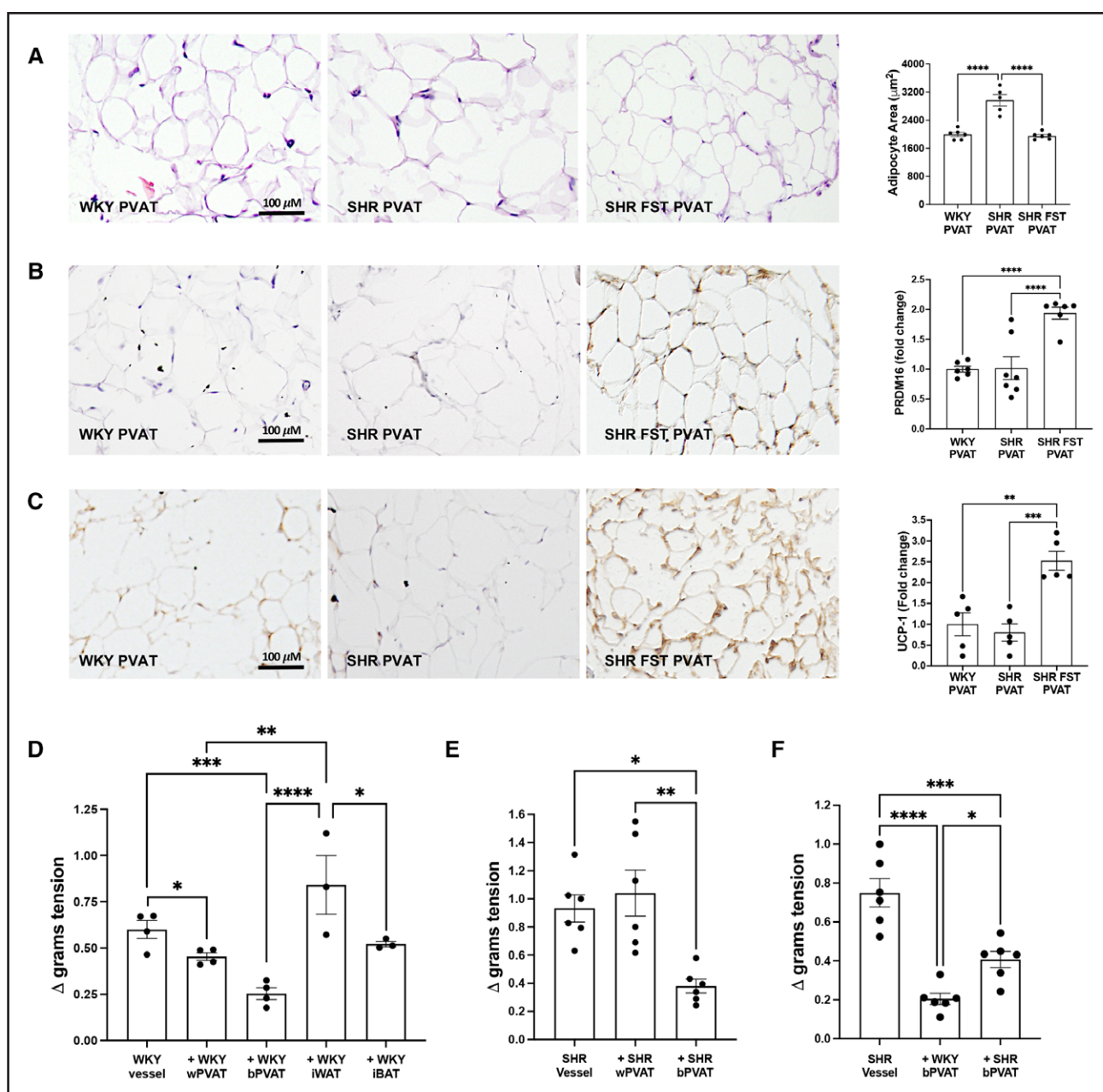


Figure 4. Follistatin (FST) induces characteristics of brown adipose tissue associated with vasorelaxant properties.

Representative cross-sectional images of mesenteric perivascular adipose tissue (PVAT) from chronically treated rats stained for (A) hematoxylin and eosin, (B) PRDM16, and (C) UCP-1, quantified in accompanying graphs. D, Response to maximal KCl dose (125 mmol/L) in first-order Wistar Kyoto (WKY) mesenteric vessels preincubated with either detached WKY white mesenteric PVAT (wPVAT), brown thoracic aorta PVAT (bPVAT), inguinal WAT (iWAT), or interscapular BAT (iBAT) for 30 minutes. E, Response to maximal KCl dose (125 mmol/L) in first-order spontaneously hypertensive rat (SHR) mesenteric vessels preincubated with either detached SHR wPVAT or bPVAT. F, Response to maximal KCl dose (125 mmol/L) in first-order SHR mesenteric vessels preincubated with either detached WKY bPVAT or SHR bPVAT. Contractility values are from 3 to 5 rats per group, 1 to 3 vessels from each rat. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

anticontractile effect in WKY vessels compared with white PVAT. Interestingly, inguinal WAT from a peripheral depot did not modulate contraction, highlighting an important functional difference between peripheral depots and PVAT. We next determined whether SHR brown PVAT showed similar impaired anticontractile effects as white PVAT. We performed detached studies in which brown or white SHR PVAT was detached and

coincubated with an SHR vessel without intact PVAT. As seen in Figure 4E, detached SHR brown but not white PVAT resulted in an anticontractile effect on SHR vessels, suggesting that a phenotypically brown PVAT may improve vascular contraction. SHR vessels demonstrated a more pronounced decrease in contraction on coincubation with brown PVAT from WKY compared with SHR (Figure 4F). These findings suggest improved regulation

of vascular function with brown PVAT from nonhypertensive animals. Together, these results illustrate potent vasodilatory effects of phenotypically brown PVAT, which is reduced in the SHR.

AMPK Regulates Follistatin-Mediated PVAT Browning

In vivo, follistatin induces browning of peripheral adipose tissue depots via increased phosphorylation and activation of ERK (extracellular signal-regulated kinase) 1/2,¹² pp38 MAPK (p38 Mitogen-activated protein kinase),¹³ and AMPK.¹⁴ Here, we demonstrate that follistatin did not affect activation of ERK 1/2 (Figure S6A), whereas phosphorylation of AMPK (Figure 5A) and pp38 MAPK (Figure S6B) were increased in SHR PVAT.

To further study the mechanism by which follistatin induces PVAT browning, we established an *ex vivo* culture system for PVAT from a protocol described by Blumenfeld et al.²² Fresh white mesenteric PVAT was isolated and cultured for 3 days. We first confirmed that browning could be established in this system by treating isolated PVAT with the β_3 adrenergic receptor agonist CL-316,243, a known inducer of browning.²³ CL-316,243 and follistatin increased PRDM16 expression in WKY PVAT (Figure 5B). *Ex vivo* 3-day follistatin treatment in SHR PVAT increased phosphorylated AMPK, which was inhibited by cotreatment with the AMPK inhibitor, compound C (Figure 5C). Follistatin also induced browning, evidenced by increased PRDM16 expression (Figure 5D). This was inhibited by compound C, suggesting that follistatin-induced PVAT browning in the SHR is dependent on AMPK.

We next determined the role of AMPK in PVAT regulation of vascular function. As shown in Figure 5E, treatment of WKY PVAT with compound C for 3 days before incubation with vessels for myography abolished its anticontractile effect to KCl. Conversely, *ex vivo* treatment of SHR PVAT with follistatin improved its ability to induce vasodilation in WKY vessels, an effect which was eliminated by cotreatment of SHR PVAT with compound C (Figure 5F). These results highlight a role for AMPK in regulating the anticontractile effects of PVAT.

We sought to quantitatively assess the association between the degree of PVAT browning and anticontractile effects. We plotted the KCl response of WKY vessels incubated with SHR PVAT treated *ex vivo* with either HBSS (control) or follistatin against the degree of browning of SHR PVAT (% positive PRDM16 stain). A higher degree of PRDM16 staining is associated with decreased vasoconstriction (Figure S7). Taken together, these results suggest that follistatin induces PVAT browning in SHRs through AMPK activation, which contributes to its anticontractile effect on vessels.

Activin A or B Neutralization Induces PVAT Browning

As follistatin is a potent inhibitor of activins A and B, we next assessed the involvement of each of these in PVAT browning using neutralizing antibodies. Activin A or B neutralization each induced browning in SHR PVAT as evidenced by increased PRDM16 expression (Figure 6A). On the contrary, treating WKY PVAT with recombinant activin A or B reduced PRDM16 expression (Figure 6B). We next sought to determine whether activin inhibition in SHR PVAT could restore its vasodilatory effects. SHR PVAT treated *ex vivo* with activin A or B neutralizing antibodies for 3 days improved PVAT ability to reduce vasoconstriction to KCl in WKY vessels (Figure 6C). Conversely, treating WKY PVAT with activin A or B for 30 minutes resulted in a loss of the anticontractile effect (Figure S8). These results suggest that activins decrease browning and contribute to PVAT dysfunction in SHRs.

Follistatin Enhances Response to β_3 Receptor Activation



To determine whether β_3 adrenergic receptors are required for follistatin-induced browning, we used the β_3 receptor blocker, L-748,337. Follistatin-induced browning of SHR PVAT, as assessed by PRDM16 expression, was reduced by β_3 receptor inhibition (Figure 6D). β_3 adrenergic receptor inhibition alone had no effect, suggesting that PVAT browning induced by follistatin requires activation of β_3 adrenergic receptors. β_3 adrenergic receptor activation is known to stimulate lipolysis, producing free fatty acids that can upregulate and activate UCP-1.²⁴ Thus, we next determined whether follistatin induces lipolysis in SHR PVAT. The release of glycerol, a byproduct of lipolysis, was measured in media collected from cultured PVAT. Activin A or B significantly reduced glycerol released by PVAT in cultured media from WKY PVAT, indicative of suppressed lipolysis (Figure 6E). Conversely, activin A or B neutralization or follistatin significantly increased lipolysis in SHR PVAT (Figure 6F). Follistatin effects were reduced by AMPK inhibition. These results demonstrate that follistatin induces lipolysis in SHR PVAT via activin inhibition and AMPK activation, suggesting an additional potential mechanism of increased browning.

We next wanted to better understand the interaction between activins, AMPK phosphorylation, and β_3 adrenergic receptors in PVAT browning. Treating SHR PVAT with activin A or B neutralizing antibodies significantly increased PRDM16 expression (Figure S9A and S9B). This effect was abolished by β_3 adrenergic receptor inhibition. To determine whether this involved AMPK, we assessed the expression of phosphorylated AMPK. Although follistatin and activin neutralization

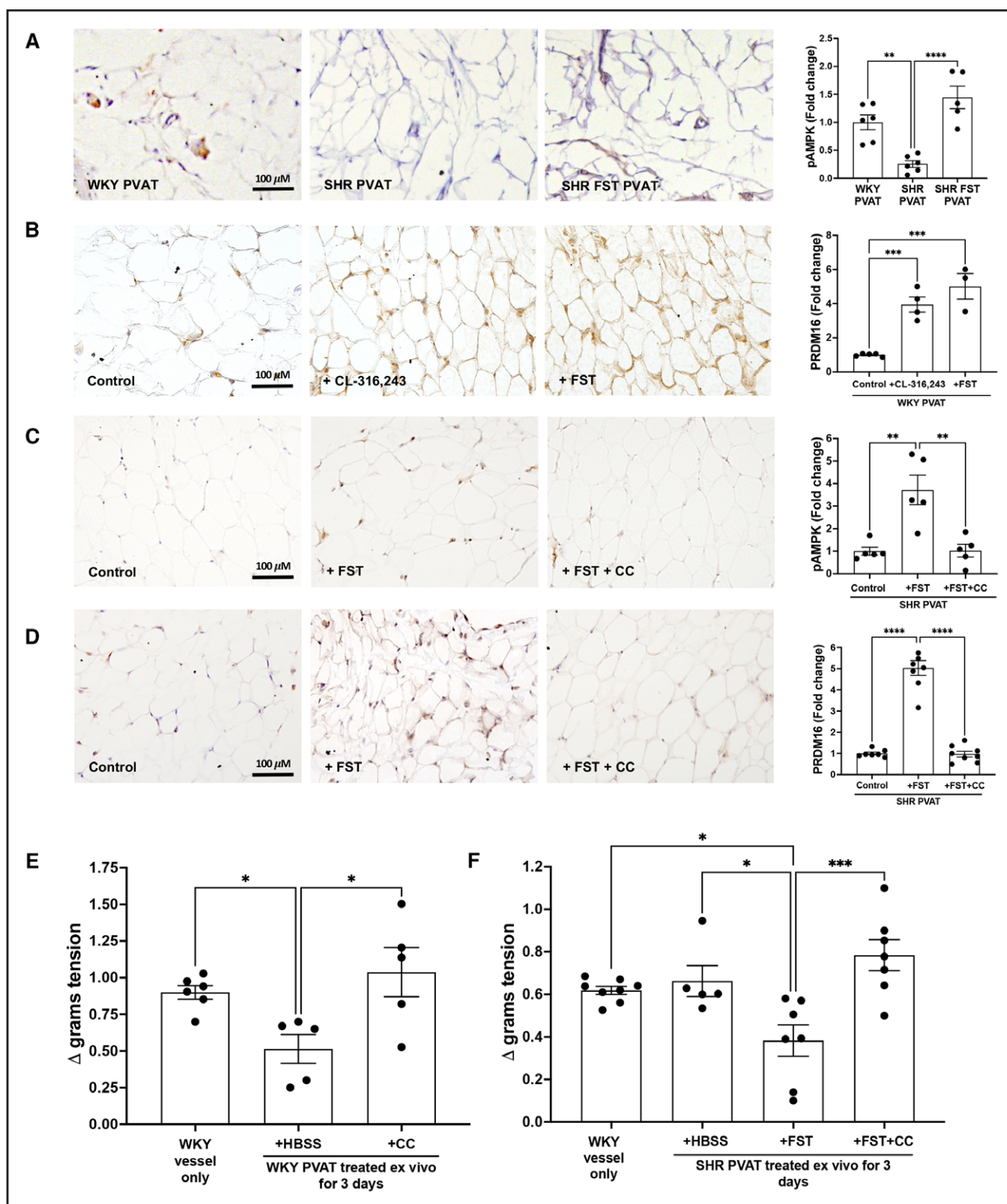


Figure 5. Follistatin (FST) induces perivascular adipose tissue (PVAT) browning via AMPK (AMP-activated protein kinase).

A, Representative cross-sectional images of mesenteric PVAT from chronically treated rats stained for phosphorylated AMPK, quantified in the accompanying graph. **B**, Representative cross-sectional images of Wistar Kyoto (WKY) mesenteric PVAT treated ex vivo for 3 days with either CL-316,243 or FST and stained for PRDM16. **C**, Representative cross-sectional images of spontaneously hypertensive rat (SHR) mesenteric PVAT treated ex vivo for 3 days with FST with or without compound C (CC) and stained for phosphorylated AMPK. **D**, Representative cross-sectional images of SHR mesenteric PVAT treated ex vivo for 3 days with FST with or without CC and stained for PRDM16. **E**, Response to maximal KCl dose (125 mmol/L) in WKY vessels incubated with mesenteric WKY PVAT treated ex vivo for 3 days with either Hank's balanced salt solution (HBSS) or CC. **F**, Response to maximal KCl dose (125 mmol/L) in WKY vessels incubated with mesenteric SHR PVAT treated ex vivo for 3 days with either HBSS, FST, or FST and CC. Contractility values are from 2 to 3 rats per group, 2 to 3 vessels per rat. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

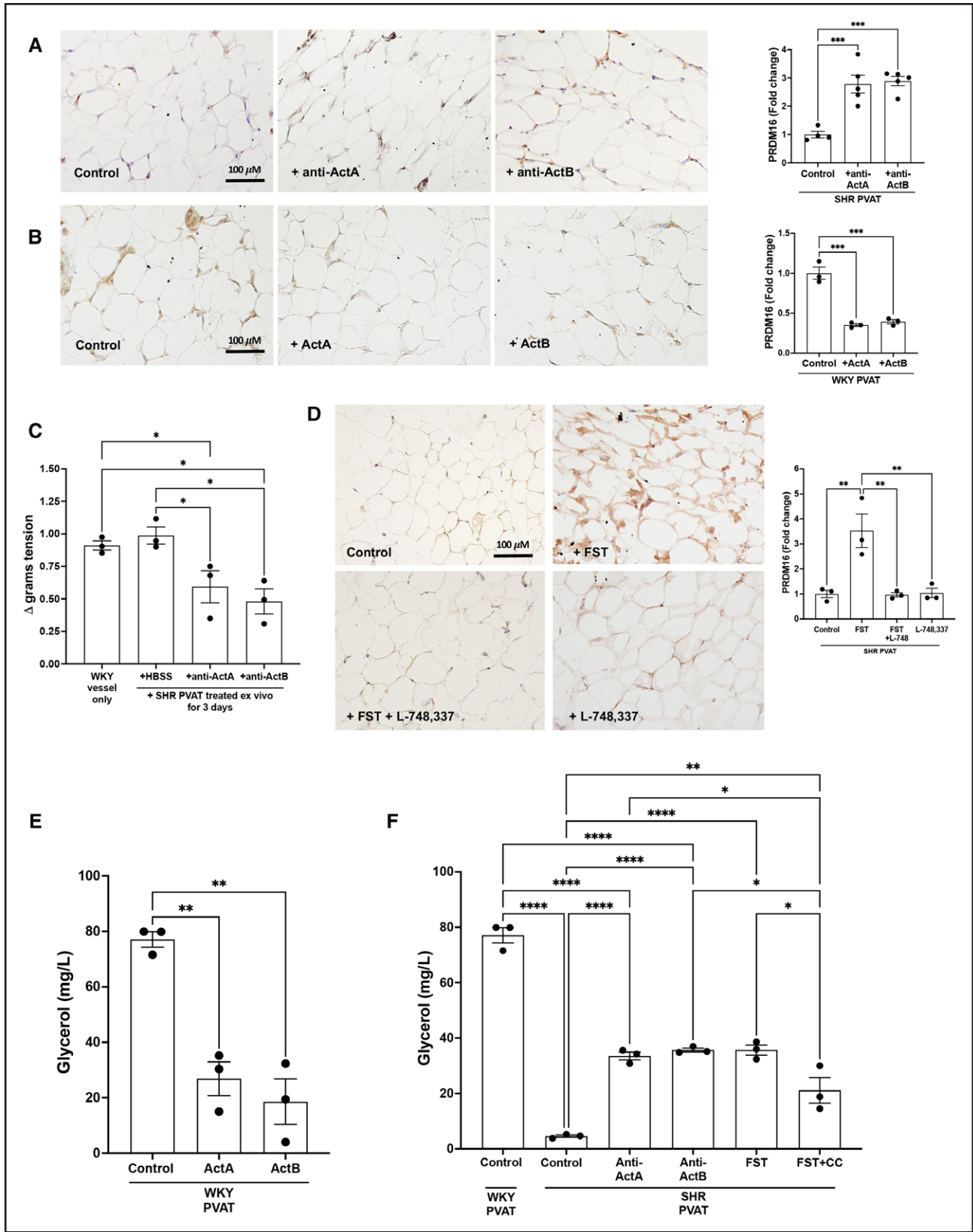


Figure 6. Follistatin (FST) induces perivascular adipose tissue (PVAT) browning and lipolysis via neutralization of activins and β_3 receptor activation. **A**, Representative cross-sectional images of spontaneously hypertensive rat (SHR) mesenteric PVAT treated ex vivo for 3 days with either Hank's balanced salt solution (HBSS) or neutralizing antibodies to activin A or activin B and stained for PRDM16, quantified in the accompanying graph. **B**, Representative cross-sectional images of Wistar Kyoto (WKY) mesenteric PVAT treated ex vivo for 3 days with either HBSS or neutralizing antibodies to activin A or activin B and stained for PRDM16, quantified in the accompanying graph. **C**, Bar graph showing Δ grams tension for WKY vessel only, +HBSS, +anti-ActA, and +anti-ActB. **D**, Representative cross-sectional images of SHR PVAT treated ex vivo for 3 days with either HBSS or neutralizing antibodies to activin A or activin B and stained for PRDM16, quantified in the accompanying graph. **E**, Bar graph showing Glycerol (mg/L) for WKY PVAT (Control, ActA, ActB). **F**, Bar graph showing Glycerol (mg/L) for WKY PVAT (Control, Anti-ActA, Anti-ActB) and SHR PVAT (Control, Anti-ActA, Anti-ActB, FST, FST+CC).

increased pAMPK, this was inhibited by β_3 adrenergic receptor inhibition (Figure S9C through S9E). To decipher how activins may regulate AMPK phosphorylation, we assessed the involvement of small mothers against decapentaplegic (Smad). SIS3, a Smad3 inhibitor, significantly increased phosphorylation of AMPK, suggesting a suppressive effect of activin-Smad signaling on AMPK (Figure S9F and S9G). These results collectively demonstrate that follistatin-induced browning requires AMPK, which is regulated by β_3 adrenergic receptors and suppressed by Smads.

The interaction between ROS, activin inhibition, and AMPK activation in promoting PVAT browning was explored. It is postulated that, in comparison to white PVAT, PVAT with a BAT-like phenotype ameliorates pathological ROS levels.²⁵ We confirm this by demonstrating that CL316-243-induced browning reduced ROS content in SHR PVAT (Figure S10A). Conversely, ROS inhibition in SHR PVAT increased PVAT browning, which was attenuated by AMPK inhibition (Figure S10B), suggesting that ROS regulates browning via AMPK. Indeed, AMPK inhibition also increased ROS in WKY PVAT (Figure S10C). Together, these results demonstrate that follistatin induces SHR PVAT browning via activin neutralization, ROS inhibition, and AMPK activation.

BP Lowering Alone Does Not Improve Vascular Function or Induce Adipose Tissue Browning

Studies have correlated adipose tissue browning with improved vascular function and lower BP.⁶ Whether reducing elevated BP results in adipose tissue browning has not been explored. Here, we assessed the effects of the vasodilator hydralazine on vascular function, targeting a similar BP reduction, which we achieved previously with follistatin. As shown in Figure S11A through S11D, BP in SHR controls (systolic blood pressure, 204 ± 15 mm Hg versus diastolic blood pressure, 149 ± 16 mm Hg), was reduced by hydralazine (systolic blood pressure, 176 ± 11 mm Hg versus diastolic blood pressure, 123 ± 19 mm Hg) to a similar extent as follistatin (systolic blood pressure, 181 ± 14 mm Hg versus diastolic blood pressure 129 ± 13 mm Hg). Unlike that seen with follistatin, however, hydralazine did not reverse the procontractile effect of SHR PVAT (Figure S11E), nor did it induce adipose tissue browning as measured by PRDM16 (Figure S11F). Altogether, BP lowering alone neither improves vascular function nor induces adipose tissue browning in

the SHR, suggesting that follistatin directly affects PVAT and vascular function to improve BP.

Follistatin Shifts the SHR AT Proteome Towards Normal State

Unbiased proteomic analysis of mesenteric PVAT from rats treated for 8 weeks with follistatin or vehicle was performed using LC-MS to investigate differential protein expression and processes regulated by follistatin compared with SHR controls (Figure S12A). A total of 3208 proteins were identified in the sample, with 2822 of these having high false discovery rate confidence. Proteins with low and medium false discovery rate confidence were not included in the analysis. An additional 64 contaminant proteins (2.88% of total) were identified in the samples and were excluded from the analysis. Using the SHR vehicle as the denominator, 24 proteins were downregulated in the SHR follistatin group, and 30 proteins were upregulated (Figure S12B). Principal component analysis plots revealed a distinguishable difference between vehicle- and follistatin-treated SHR (Figure S12C), with the direction of the follistatin-treated SHR cluster resembling that of WKY controls. Heatmap analysis of protein abundances highlights distinct profiles between vehicle- and follistatin-treated SHR (Figure S12D). Protein abundances from follistatin-treated SHR align more closely to WKY controls. Gene Ontology analysis revealed 52 biological processes significantly upregulated in PVAT by follistatin compared with SHR PVAT (Figure S12E), with 63% of these also upregulated in WKY PVAT compared with SHR PVAT. Follistatin increased processes that regulate adipose tissue browning, including cAMP and PKA signaling. Collectively, these data reveal that follistatin induces a shift in the SHR PVAT proteome toward a normal PVAT profile.

DISCUSSION

Impairment in PVAT, which normally functions to exert an overall anticontractile effect, is increasingly recognized as an important contributor to hypertension. In this study, we describe the dysregulation of vascular tone by SHR PVAT. Follistatin restores PVAT function by inhibiting oxidative stress and inducing browning, mediated by neutralization of activins A and B and AMPK activation. This is the first study to assess the effects of PVAT oxidative stress and browning on resistance artery function

Figure 6 Continued. 3 days with either HBSS, activin A, or activin B and stained for PRDM16. **C**, Response to maximal KCl dose (125 mmol/L) in WKY vessels incubated with mesenteric SHR PVAT treated ex vivo for 3 days with either HBSS or neutralizing antibodies to activin A or activin B. **D**, Representative cross-sectional images of SHR mesenteric PVAT treated ex vivo for 3 days with either HBSS, FST, FST and L-748 337, or L-748 337 alone, quantified in the accompanying graph. **E**, Glycerol content measured in media from WKY PVAT treated with either HBSS, activin A, or activin B for 24 hours. **F**, Glycerol content measured in media from SHR PVAT treated with either HBSS, neutralizing antibodies to activin A or activin B, FST, or FST and compound C (CC) for 24 hours. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

in hypertension, identifying follistatin as a novel regulator restoring PVAT function.

We had previously identified that SHR vessels exhibit increased oxidative stress, and this is reduced by follistatin in association with improved vessel function.⁷ Although the mesenteric PVAT anticontractile effect is known to be impaired in this model,²⁶ the effect of follistatin on PVAT was unknown. We now show that the increased oxidative stress seen in SHR PVAT, known to contribute to its dysfunction,²⁷ is abolished by both chronic *in vivo* and acute *ex vivo* follistatin treatment. This reduction in oxidative stress by follistatin has also been observed in other settings, such as chronic kidney disease, with reduced kidney mass.²⁸ Intriguingly, our data further highlight the ability of PVAT to induce oxidative stress in a closely associated vessel, an effect which is prevented by follistatin. Indeed, the ability of PVAT-generated ROS to diffuse towards vascular cells and cause oxidative damage and impaired vessel function is known.²⁹ Follistatin thus has dual protective effects on the vessel in hypertensive animals, which are both direct and indirect through reducing PVAT oxidative stress.

Follistatin neutralizes several members of the TGF β superfamily, with greatest potency against activins A and B. We and others have reported that activin A contributes to vessel ROS production in both systemic and pregnancy-associated hypertension.^{7,9} However, the contribution of activin B to oxidative stress in general, as well as the contribution of activins A and B to PVAT oxidative stress and dysfunction in hypertension, were unknown. We demonstrate increased abundance of activins A and B in SHR PVAT, both of which were reduced by follistatin. Activin A or B treatment of normal PVAT augmented ROS production, suggesting that activin signaling directly contributes to ROS production in PVAT. In vessels, ROS production in response to activin A was mediated by its induction of NOX.⁹ Here, we demonstrate that activin A also produces ROS through NOX in PVAT, suggesting that follistatin may lower NOX-induced oxidative stress via activin neutralization.

ROS contribute to hypertension through reduction in bioavailability of the vasodilator, NO.³⁰ PVAT is a major producer of NO, which is known to be reduced in SHR PVAT.²⁷ Here we show that follistatin improves NO bioavailability in SHR PVAT, mediated by both activins A and B. Within PVAT, NO is produced primarily by eNOS. Activation of eNOS, as measured by its phosphorylation at S1177, was not different between WKY and SHR PVAT, in keeping with findings by Carvalho et al.²⁷ However, follistatin increased eNOS activation in SHR PVAT, suggesting that the restored NO bioavailability may be due to both reduced scavenging by superoxide as well as increased production of NO.

The presence of BAT in individuals correlates with lower prevalence of hypertension,⁵ suggesting that increased browning of PVAT may promote vasoprotective

effects. Indeed, ablation of beige fat and browning results in vascular and PVAT remodeling and elevated BP.³¹ Conversely, β_3 agonist-induced browning of PVAT increases its anticontractile effect³⁰ and lowers BP.⁶ Here we show that follistatin induces browning of both peripheral adipose depots, as seen previously,^{12–14} and PVAT. Furthermore, we show that brown PVAT induces a more potent anticontractile effect compared with white PVAT. We additionally highlight a functional difference between peripheral adipose tissue depots and PVAT, such that PVAT is more anticontractile than peripheral depots, even within the same phenotype. Heterogeneity in the profiles of secreted vasoactive factors and activity of the sympathetic nervous system are likely contributors to these differences.

While follistatin did not alter ERK activation in our study, both p38 MAPK and AMPK activation were reduced in SHR PVAT and increased by follistatin. As AMPK deficiency results in suppressed PVAT NO synthesis and impaired anticontractile activity,³² we elected to focus further on its involvement. We identified that follistatin-induced PVAT browning and its anticontractile properties are dependent on AMPK activity through inhibition of activin-Smad signaling. Suppression of Smad activation by AMPK has also been demonstrated in adipocytes³³ proposing a potential bidirectional regulation. Notably, a similar AMPK-dependent browning phenotype is seen in mice deficient in myostatin, another member of the TGF β superfamily inhibited by follistatin.³⁴ Further studies should be conducted to assess whether myostatin inhibition may have contributed to our findings.

We demonstrate the requirement for β_3 adrenergic receptors in the phosphorylation of AMPK in SHR PVAT. β_3 adrenergic receptor activation is an important stimulator of browning through its induction of lipolysis and consequent release of free fatty acids that bind to and activate UCP-1. Activin B and Smad2/3 signaling were shown to inhibit lipolysis in adipocytes,³⁵ suggesting that activins may suppress browning. Supporting this finding, we discovered reduced lipolytic activity in SHR PVAT, which was improved with follistatin and neutralization of both activin A and B. The improved lipolysis with follistatin was reduced by inhibition of AMPK, a well-known mediator of β_3 adrenergic effects.³⁶ In line with this, β_3 adrenergic receptor antagonism resulted in a significant reduction in AMPK phosphorylation. These findings were observed in the absence of a receptor agonist, suggesting constitutive β_3 adrenergic receptor activity. Indeed, constitutive activity of this receptor was observed in cultured 3T3-F442A adipocytes.³⁷ Due to the lack of a specific inverse agonist of β_3 adrenergic receptors, this has not been determined directly. Collectively, our results suggest that follistatin induces browning through the inhibition of activin-Smad signaling, resulting in the increased phosphorylation of AMPK, which is regulated basally by β_3 adrenergic receptors.

In line with data demonstrating reduced oxidative stress on browning,³⁸ we demonstrate reduced ROS on induction of browning in SHR PVAT. Conversely, reduction in BAT through UCP-1 deficiency results in increased oxidative stress and elevated BP,³⁹ associated with reduced AMPK activation.⁴⁰ Our findings showing that AMPK inhibition increases PVAT ROS in association with a reduced brown phenotype suggest that AMPK may play a central role in regulating thermogenic and redox balance in PVAT.

Proteomic analysis of mesenteric PVAT broadly showed that follistatin shifted the proteome toward characteristics of normal PVAT. Pathway enrichment analysis identified processes associated with cytokine secretion, cell cycle regulation, extracellular matrix, immune response, and prostaglandin synthesis as being restored by follistatin, in keeping with the multicellular composition and sophisticated environment that constitutes PVAT. While these data support mechanisms regulating browning, including cAMP and PKA signaling, future studies would assess additional processes in more detail. The following are worth noting: (1) Follistatin increased synthesis of prostaglandins, which may induce vasodilation, such as those signaling through E-prostanoid receptors 2 and 4 (EP₂, EP₄).⁴¹ Indeed, EP₂-induced vasodilation is known to be reduced in SHR.⁴² Furthermore, the enzyme cyclooxygenase, a proximal regulator of prostaglandin synthesis, is required for cold-induced browning of inguinal WAT.⁴³ (2) Increased apoptosis also characterizes PVAT dysfunction in the SHR.²⁷ One of the most upregulated proteins by follistatin was the anti-apoptotic protein CIAPIN1 (cytokine-induced apoptosis inhibitor 1), suggesting that follistatin may also protect against this process.

This study has several limitations. First, only male animals were included in the analysis, and therefore potential sex-specific differences cannot be assessed. Given known sex differences in hypertension, future studies incorporating female animals will be important to determine the generalizability of these findings. Second, PVAT is heterogeneous, producing a wide range of bioactive mediators. Proteomic analysis would not capture nonprotein factors such as gaseous mediators or lipid-derived signaling molecules, which may contribute to vascular regulation. Our proteomic analysis is inherently limited by the availability and annotation of characterized proteins within the rat proteome database, which may restrict the identification of relevant targets. Further, we did not characterize the PVAT secretome to identify specific PVAT-derived factors that may contribute to its anticontractile effect. Lastly, we cannot ignore the potential influence of renal and metabolic changes influenced by chronic follistatin treatment on PVAT phenotype in the SHRs.

In summary, our findings establish follistatin as a potent modulator of PVAT function, capable of preventing its procontractile effect through activin neutralization,

AMPK-dependent browning, and attenuation of oxidative stress. The improved vascular function is associated with lower BP in the SHR model of essential hypertension. These insights underscore the role of PVAT in vascular homeostasis, advancing our understanding of adipose-vascular crosstalk. Our findings provide a mechanistic foundation for the development of PVAT-targeted therapies to mitigate vascular dysfunction in essential hypertension.

PERSPECTIVES

Essential hypertension is a leading contributor to cardiovascular morbidity, yet current therapies do not sufficiently control BP in a third of hypertensive patients. Furthermore, their ability to reverse maladaptive vascular structural and functional changes is variable, with some classes having no effect. PVAT serves as a critical regulator of vascular homeostasis, exerting a baseline anticontractile effect that promotes arterial relaxation. However, the transition to a dysfunctional PVAT phenotype, characterized by a loss of these vaso-protective properties, is a major factor in the pathogenesis of systemic hypertension. Our study confirms significant PVAT dysfunction in a model of essential hypertension, characterized by enhanced oxidative stress, vascular contractility, and levels of the cytokines activins A and B. The activin antagonist follistatin reduces PVAT oxidative stress and restores its anticontractile properties, associated with a reduction in systemic BP. Importantly, follistatin also promotes PVAT browning via activin inhibition and increased AMPK activity. Notably, BP reduction alone with hydralazine fails to rescue the PVAT phenotype, indicating that follistatin-induced improvements are independent of systemic hemodynamics. By delineating the molecular pathways through which follistatin modulates PVAT health, this work advances our understanding of adipose-vascular crosstalk and introduces a novel strategy for mitigating vascular dysfunction and improving BP in essential hypertension.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Expanded Supplementary Methods

Figures S1–S12

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